

Knapack Problem Greedy Method

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Made with KINEMASTER
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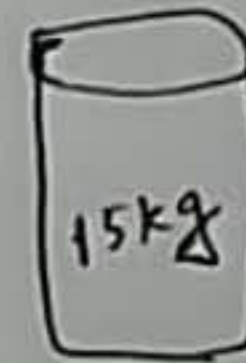
Objects:	1	2	3	4	5	6	7
Profit(P):	12	5	16	7	9	11	6
Weight(W):	3	1	4	2	9	4	3

$n = 7$

$m = 15$

$\max(\frac{P}{W})$

4	5	4	3.5	1	2.75	2
↑	↑	↑	↑		↑	↑



Object n	Profit	Weight	Remaining Weight
2	5	1	$15 - 1 = 14$
1	12	3	$14 - 3 = 11$
3	16	4	$11 - 4 = 7$
4	7	2	$7 - 2 = 5$

$\frac{\text{max profit}}{43}$

$\frac{\text{min profit}}{(27)}$

When solving by profit we use the order by profit and not the ratio

Weight (w): 3 1 4 2 9 4 3 | m = 15

max ($\frac{p}{w}$):

4	5	4	3.5	1	2.75	2
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Object n	Profit n	Weight	Remaining Weight
2	5	1	$15 - 1 = 14$
1	12	3	$14 - 3 = 11$
3	16	4	$11 - 4 = 7$
4	7	2	$7 - 2 = 5$
6	11	4	$5 - 4 = 1$
7	$6 \times \frac{1}{3} = 2$	1	$1 - 1 = 0$
56		15	

max profit
43

min profit
(27)

or, $2 \times 1 = 2$